

Shanmukha Chatadi

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SUMMARY

AI/ML Engineer with production experience architecting RAG platforms, retrieval pipelines, and ML systems on AWS at enterprise scale. Published researcher in deep learning for remote sensing and medical imaging. MS CS from NC State (Data Science), skilled in Python, LLMs (Claude, Gemini), LangChain, and cloud-native ML deployment.

EDUCATION

North Carolina State University Master of Science in Computer Science, GPA: 3.71/4.00	Raleigh, NC Aug 2024 – May 2026
National Institute of Technology Puducherry Bachelor of Technology in Computer Science and Engineering, GPA: 8.48/10.00	Pondicherry, India Dec 2020 – Mar 2024

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript, SQL, Go, Java, C++ , Bash
AI/ML	LLMs, RAG, Agentic AI, NLP, Prompt Engineering, Embeddings, Fine-tuning, MLOps, Model Evaluation
Frameworks	PyTorch, TensorFlow, Hugging Face Transformers, LangChain, LangGraph, LlamaIndex, FastMCP
Backend & Cloud	FastAPI, Flask, Node.js, React, Next.js, REST APIs, GraphQL, Microservices, AWS (Bedrock, Lambda, S3, API Gateway, Cognito, OpenSearch), Docker, Kubernetes, Terraform, CI/CD
Data & Storage	PostgreSQL, MongoDB, Redis, ChromaDB, Pinecone, OpenSearch (k-NN), Elasticsearch
Tools & Infra	Git, Linux, GitHub Actions, Jenkins, Prometheus, Grafana, pytest, Jira, Agile/Scrum

WORK EXPERIENCE

AI Research Intern, Michigan Health Information Network Shared Services (MiHIN), Lansing, MI • Architected a serverless RAG platform on AWS (Bedrock, Lambda, OpenSearch Serverless, Cognito) for 230+ employees to query 10K+ enterprise documents ; pitched it to C-suite and won org-wide pilot approval. • Built agentic, tool-calling workflows with the AWS Strands Agents SDK , orchestrating Claude over MCP tool servers to run multi-step tasks autonomously instead of single-shot Q&A. • Engineered the NLP retrieval pipeline (parsing, Titan V2 embeddings, configurable chunking, Claude Sonnet) and tuned retrieval-K with human-in-the-loop feedback, cutting ungrounded LLM responses by 30% . • Integrated AI workflows behind secure REST APIs (API Gateway, Lambda, Cognito) with structured logging, CloudWatch alarms, and dead-letter routing, cutting mean time to detection by 60% .	Aug 2025 – Present
Software Engineer, Rygen BioPharma, Hyderabad, India • Built a clinical data platform (Flask, React, PostgreSQL) with 15+ REST endpoints for 4+ research teams, replacing spreadsheet-driven workflows and cutting reporting overhead by 40% . • Designed Python ETL pipelines (Pandas, NumPy) processing 100K+ clinical records daily , with PostgreSQL trigger-based validation that reduced data entry errors by 50% . • Containerized services with Docker and deployed via Kubernetes and Jenkins CI/CD; authored 200+ unit and integration tests at 85%+ coverage, shrinking release cycles from 2 days to under 3 hours . • Automated weekly R&D reporting with scikit-learn and Power BI, replacing manual Excel pipelines and freeing researchers 10+ hours/week for bench work.	Mar 2023 – Aug 2024
Data Science Intern, TreoSoft IT Solutions, Bengaluru, India • Developed an FP-Growth recommendation engine mining association rules over 100K+ retail transactions; deployed as a Flask REST API with PostgreSQL, contributing to a 15% net sales lift . • Performed EDA (Pandas, Matplotlib) to surface co-purchase patterns and designed price-anchored bundle strategies for merchandising rollout.	Jun 2022 – Sep 2022

PROJECTS

ScholarAgent <i>LangGraph, ChromaDB, FastAPI, Next.js 14, TypeScript</i> [GitHub] • Orchestrated a 7-node agentic RAG pipeline in LangGraph, routing, retrieval, relevance grading, query rewriting, generation, hallucination detection, and synthesis implementing corrective RAG that improved answer groundedness by 35% over 10K+ arXiv/PubMed papers. • Integrated ChromaDB vector search with sentence-transformer embeddings, a 3-model Gemini cascade for rate-limit resilience, WebSocket step streaming, and 21+ backend tests with GitHub Actions CI.	
MCP Healthcare Server <i>Python, FastMCP, httpx, openFDA/PubMed/NLM/CMS APIs, Docker</i> [GitHub] • Built an Anthropic Model Context Protocol server exposing 6 healthcare tools (drug safety, ICD-10 lookup, clinical literature, Medicare data, drug interactions) callable from Claude; integrated 4 federal APIs with async httpx at sub-second latency. • Implemented fuzzy ICD-10 search over 70K+ codes via NLM Clinical Tables API; containerized with Docker and wrote 21 unit tests with mocked HTTP across Python 3.11–3.13.	

PUBLICATIONS

C. Shanmukha Srinivas Sai et al., Potato Disease Classification Using ML Models, HIS 2023 (Springer) **Somesh K, C. Shanmukha Srinivas Sai et al., Tea Leaf Disease Classification with Ensemble Stacking**, ICDLAIR 2024 (IEEE)